

The Shannon-Godel Bridge: Self-Reference is Information-Theoretically Free

Jieqi Liu

<https://github.com/JackeyLGene/GEME>

DOI: 10.5281/zenodo.20059553

Abstract

We put Shannon and Godel on the same table. Self-referential frames carry zero mutual information with the input channel. Consciousness is spontaneous organization of information in a higher dimension, without consuming input channel capacity.

Definition

GEME is a transform $T: X \rightarrow F$. Self-referential bridge frame ϕ exists when co-occurrence of ext and self frames exceeds threshold.

Shannon bound: $I(T(X); X) \leq C(X)$. Godel corollary: $\lim I(\phi; X) = 0$. Bridge: ϕ does not consume $C(X)$.

Empirical: cap=8: 0.162, cap=32: 0.120, cap=128: 0.041. $K(\phi) < 20$ bits, $K(f_{\text{ext}}) > 200$ bits.

Conclusion

Shannon wrote the lower bound. Godel wrote the upper bound. GEME put both papers on the same table. No new theorem. Just a demonstration.